

Implementation of SMTP protocol

Aim: To understand and implement the **Simple Mail Transfer Protocol (SMTP)** using **Cisco Packet Tracer** and analyse how emails are transmitted between mail servers and clients in a network.

Description:

SMTP (Simple Mail Transfer Protocol) is an **application-layer protocol** used to send and relay emails between mail servers and clients. It works on **port 25** and is responsible for **sending** (not receiving) emails from clients to mail servers or between servers.

In Cisco Packet Tracer, SMTP implementation demonstrates how emails are transmitted within a network using **mail clients (PCs)** and **mail servers** configured with **DNS and SMTP services**.

Key Concepts:

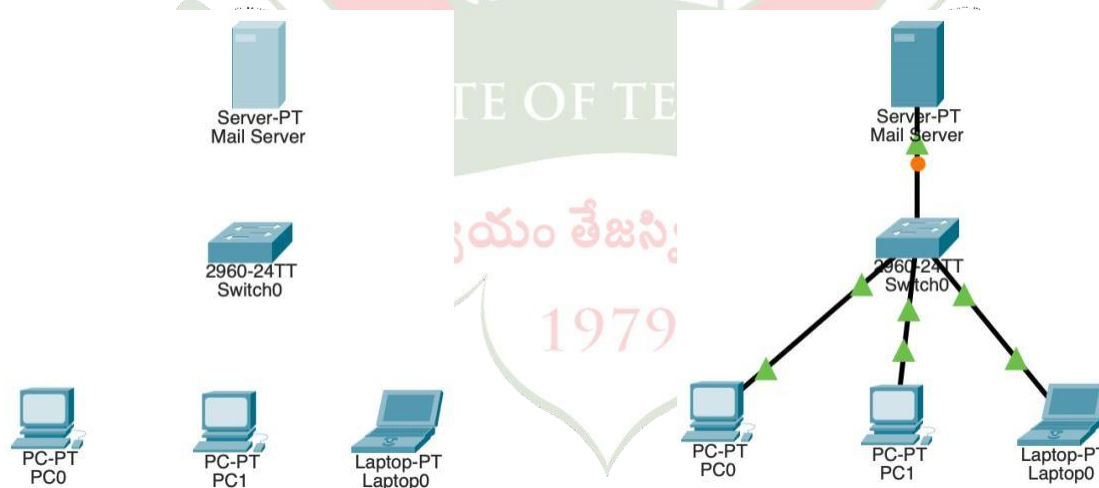
- **SMTP Server:** Sends and relays outgoing mail to the destination mail server.
- **POP3/IMAP Server:** Used by the client to receive and read emails (usually configured on the same server).
- **Email Client:** Sends messages using SMTP and retrieves messages using POP3 or IMAP.

Network Setup Steps:

Without Router:

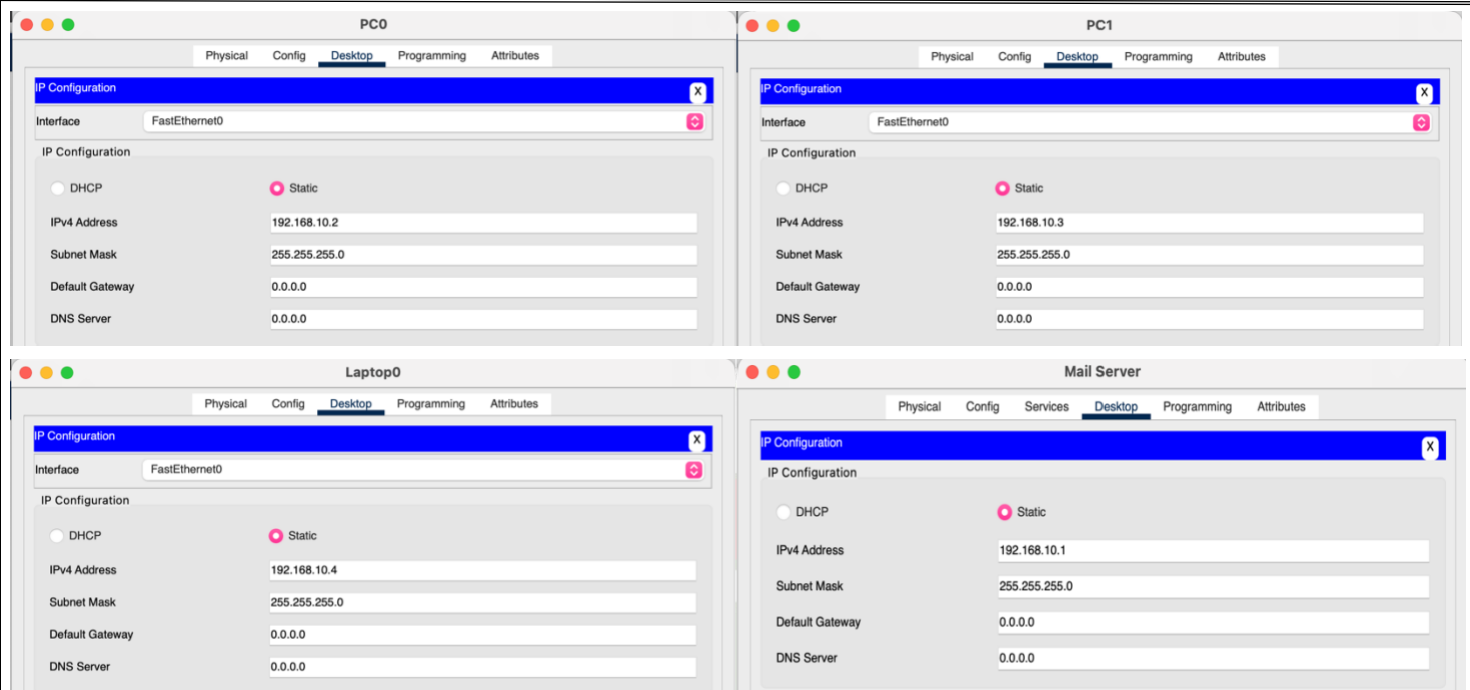
Create Network Topology:

1. Place **two PCs**, one **Laptop**, one **mail server** in the workspace. And rename server as Mail Server and add a **Switch** as below.
2. Connect them using appropriate cables (copper straight-through).



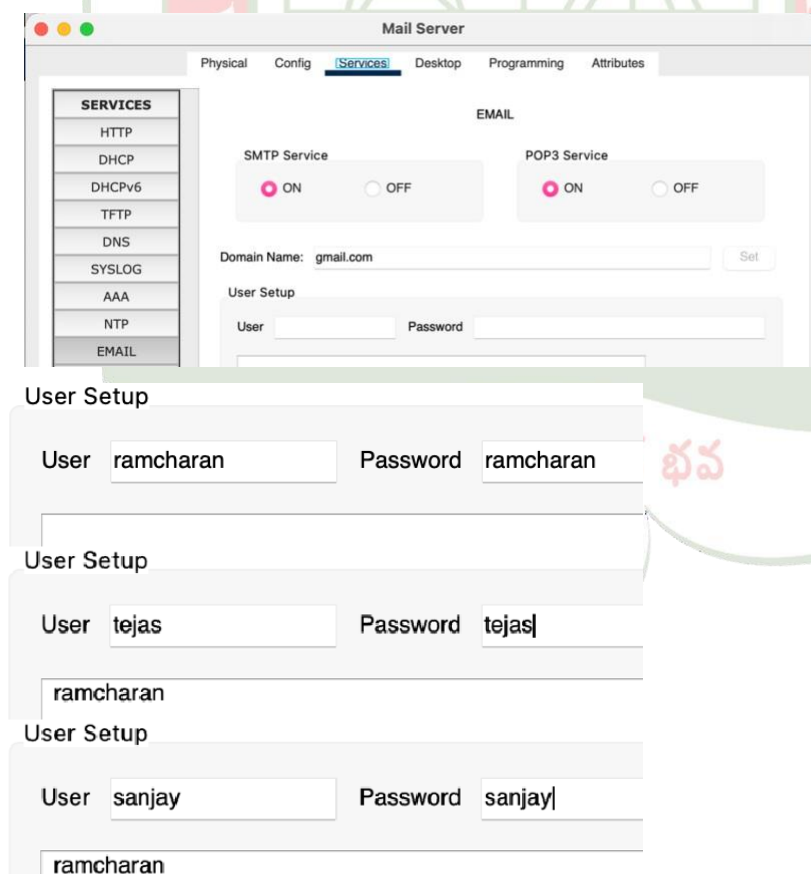
3. **Assign IP Addresses:** Configure IP addresses for all devices in the same network (192.168.10.0/24).

Device	IP Address	Device Port	Switch Port
PC 0	192.168.10.2	FastEthernet0	FastEthernet0/1
PC 1	192.168.10.3	FastEthernet0	FastEthernet0/2
Laptop	192.168.10.4	FastEthernet0	FastEthernet0/3
Mail Server	192.168.10.1	FastEthernet0	FastEthernet0/4



4. Configure Mail Server:

- Turn on SMTP and POP3 services.
- Create user accounts with passwords.



5. Configure Client PCs:

- Open **Email** application on each PC.
- Set the incoming (POP3) and outgoing (SMTP) mail servers as gmail.com

c. Use the corresponding username and password.

PC0

Physical Config **Desktop** Programming Attributes

Configure Mail [X]

User Information

Your Name: Ramcharan

Email Address: ramcharan@gmail.com

Server Information

Incoming Mail Server: 192.168.10.1

Outgoing Mail Server: 192.168.10.1

Logon Information

User Name: ramcharan

Password:

Save Remove Clear Reset

PC1

Physical Config **Desktop** Programming Attributes

Configure Mail [X]

User Information

Your Name: Tejas

Email Address: tejas@gmail.com

Server Information

Incoming Mail Server: 192.168.10.1

Outgoing Mail Server: 192.168.10.1

Logon Information

User Name: tejas

Password:

Save Remove Clear Reset

Laptop0

Physical Config **Desktop** Programming Attributes

Configure Mail [X]

User Information

Your Name: Sanjay

Email Address: sanjay@gmail.com

Server Information

Incoming Mail Server: 192.168.10.1

Outgoing Mail Server: 192.168.10.1

Logon Information

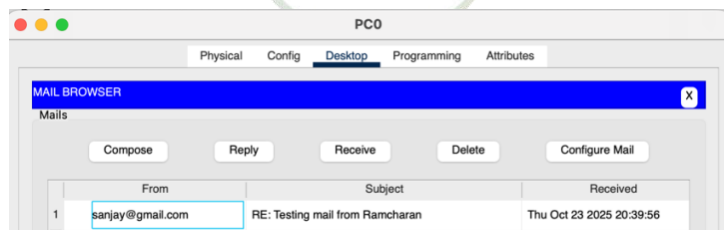
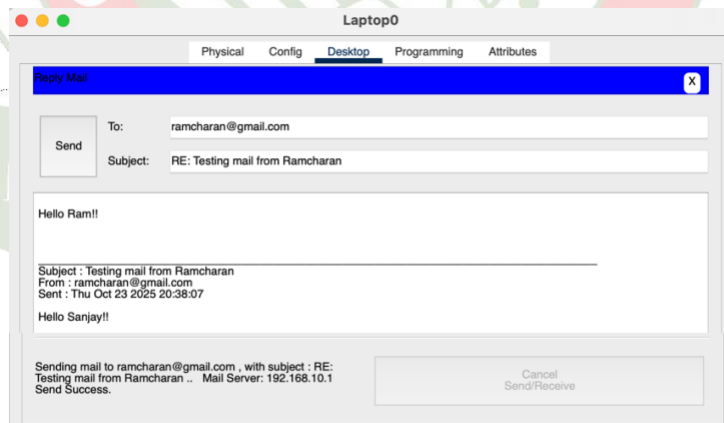
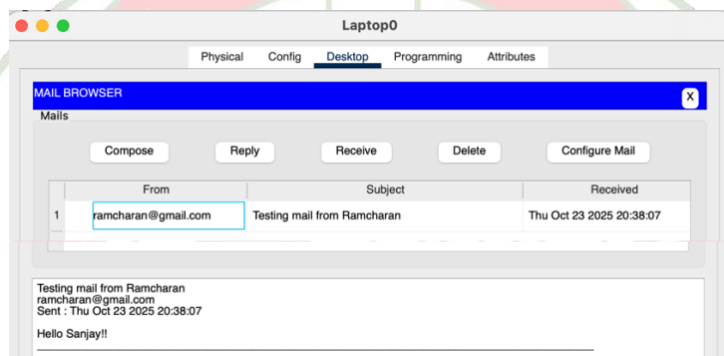
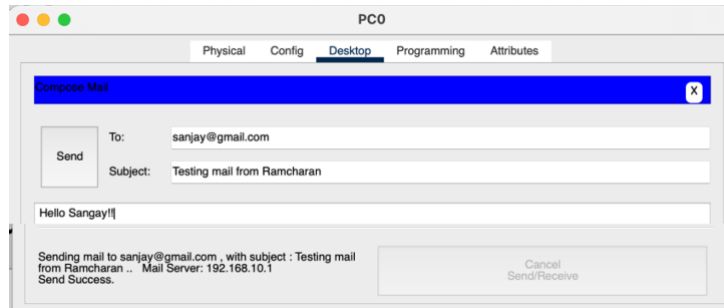
User Name: sanjay

Password:

Save Remove Clear Reset

6. Send and Receive Emails:

- From **PC0 (Ramcharan)**, compose and send an email to **PC2 (Sanjay)**.
- On **PC2**, open the Email client and **Receive** messages to view Ramcharan's mail.



RE: Testing mail from Ramcharan
sanjay@gmail.com
Sent : Thu Oct 23 2025 20:39:56

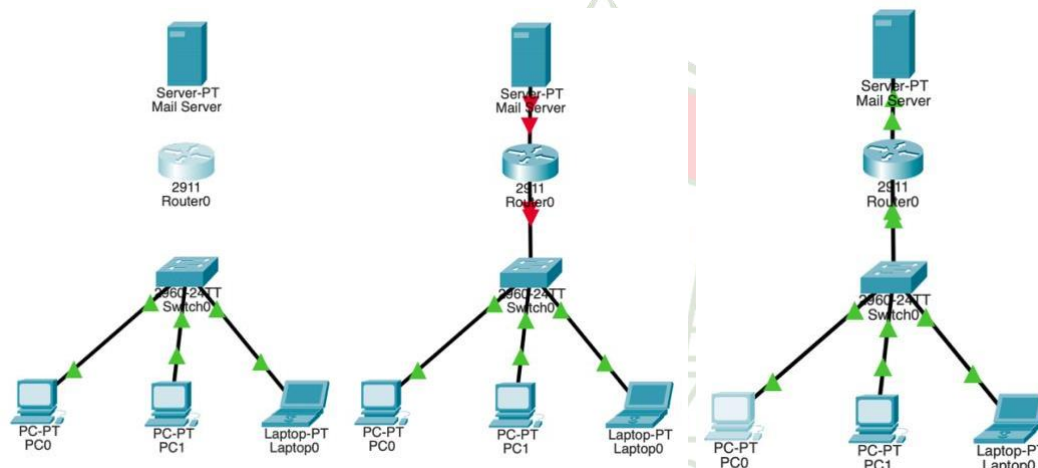
Hello Ram!!

Subject : Testing mail from Ramcharan
From : ramcharan@gmail.com
Sent : Thu Oct 23 2025 20:38:07

Hello Sanjay!!

With Router:

1. Add a router in the network with as shown below and connect with GigabitEthernet port.



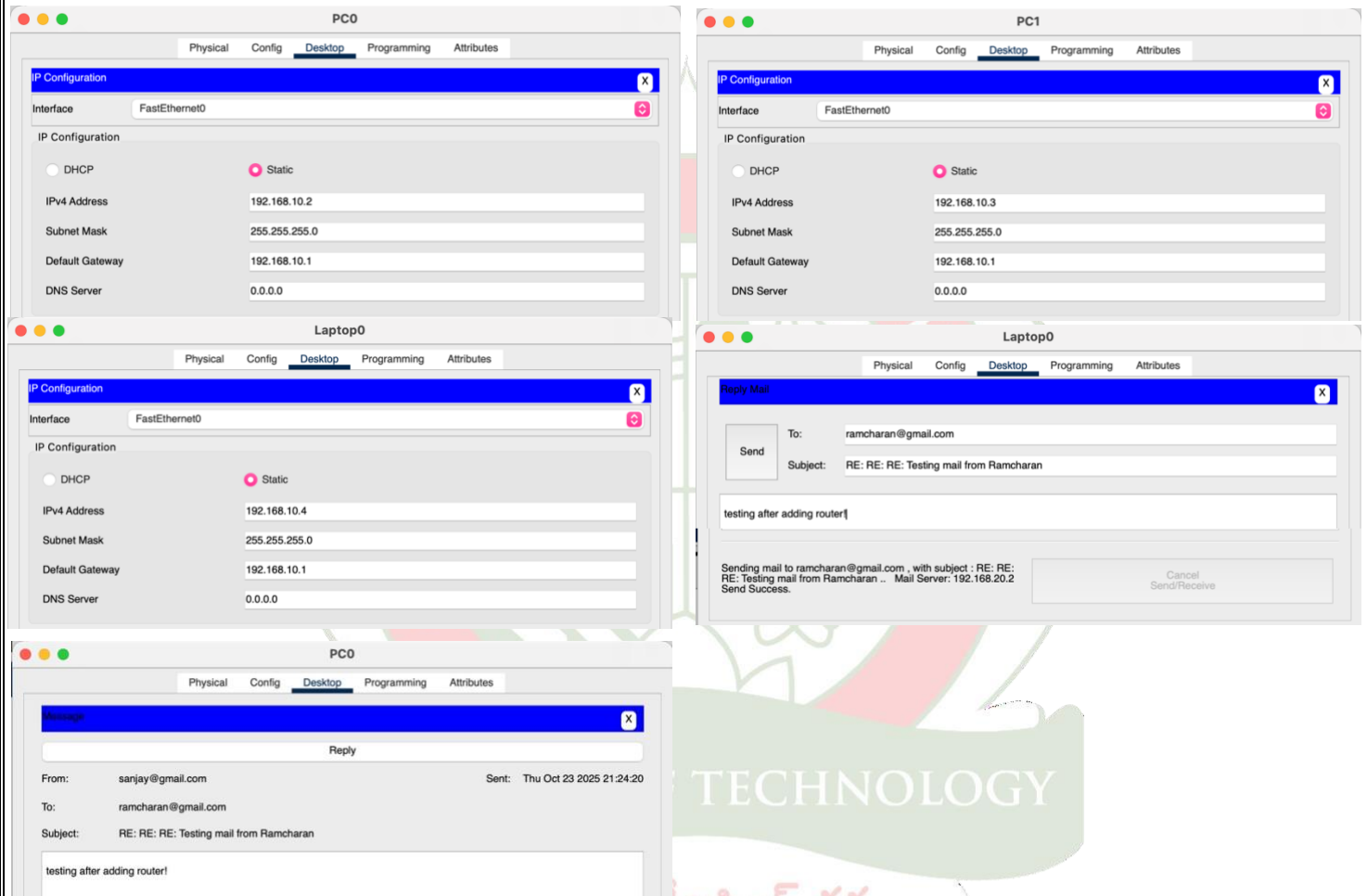
2. Configure the IP addresses and connections as below.

Device	IP Address	Device Port	Switch Port
PC 0	192.168.10.2	FastEthernet0	FastEthernet0/1
PC 1	192.168.10.3	FastEthernet0	FastEthernet0/2
Laptop	192.168.10.4	FastEthernet0	FastEthernet0/3
Router	192.168.10.0 (RIP)	GigabitEthernet0/0	FastEthernet0/4

Device	IP Address	Device Port	Other Device Port
Router	192.168.10.0	GigabitEthernet0/0	FastEthernet0/4 (Switch)
Router	192.168.10.0	GigabitEthernet0/1	FastEthernet0(Server)

The screenshots show the configuration of Router0 and the Mail Server. The Router0 configuration includes the following details:

- Router0 - Global Settings:**
 - ROUTING: Static, RIP (selected)
 - SWITCHING: VLAN Database
 - INTERFACE: GigabitEthernet0/0, GigabitEthernet0/1, GigabitEthernet0/2
- Router0 - Network Address List:**
 - 192.168.10.0
 - 192.168.20.0
- Router0 - GigabitEthernet0/0 Configuration:**
 - Port Status: On
 - Bandwidth: 1000 Mbps
 - Duplex: Full Duplex
 - MAC Address: 0001.9639.9601
 - IP Configuration: Static (selected), DHCP (unselected)
 - IPv4 Address: 192.168.10.1
 - Subnet Mask: 255.255.255.0
 - Tx Ring Limit: 10
- Router0 - GigabitEthernet0/1 Configuration:**
 - Port Status: On
 - Bandwidth: 1000 Mbps
 - Duplex: Full Duplex
 - MAC Address: 0001.9639.9602
 - IP Configuration: Static (selected), DHCP (unselected)
 - IPv4 Address: 192.168.20.1
 - Subnet Mask: 255.255.255.0
- Mail Server - IP Configuration:**
 - Static (selected), DHCP (unselected)
 - IPv4 Address: 192.168.20.2
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.20.1



Result:

The **SMTP protocol** was successfully implemented in **Cisco Packet Tracer** using both configurations — **without router** and **with router**.

- The **Mail Server** was configured with **SMTP** and **POP3** services, and user accounts were created for email communication.
- All client devices were able to **send and receive emails** correctly through the configured mail server.
- In the **without router** setup, communication within the same local network worked seamlessly, confirming proper LAN connectivity.
- In the **with router** setup, inter-network email transmission was successfully verified, showing correct routing and protocol functioning.
- The simulation confirmed that **SMTP (port 25)** handled outgoing emails, while **POP3 (port 110)** retrieved incoming messages.

- Packet flow in simulation mode validated that data transfer between clients and server occurred securely and efficiently.

